

Black Americans, Legendary Inventors and Innovators

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Abstract

Black American inventors modernized the United States of America. Labor and intellect underpinned industry and science. Innovations improved productivity, while advanced cultivation practices increased yields. Thought of as mere hammers, architects and laborers built foundational infrastructures. Lewis Latimer's U.S. Patent No. 252,386, c. 1,882 CE, turned an experimental project—the incandescent light bulb—into a revolutionary technology used in homes, streets, and businesses. A polymath, patent investigator, expert witness, and assistant to Alexander Graham Bell, Latimer drafted blueprints for the telephone. Described as "*human computers*," NASA mathematicians performed complex calculations for the moon landing.

Introduction

Black Americans attended 101+ HBCUs and registered 50,000+ patents, c. 1,800-1,900 CE. Ingenious inventions, innovations, and achievements in agriculture, mechanics, medicine, architecture, technology, science, mathematics, astronomy, and NASA's race to space moved the United States of America forward.

Black Americans, Legendary Inventors and Innovators

Lewis Howard Latimer: Lewis Latimer's U.S. Patent No. 252,386, c. 1,882 CE, turned an experimental project—the incandescent light bulb—into a revolutionary technology used in homes, streets, and businesses. A polymath, patent investigator, expert witness, and assistant to Alexander Graham Bell, Latimer drafted blueprints for the telephone.

Emmitt McHenry: Masterminded and managed the registration system for .com, .net, and .org domains, which allowed computer users to navigate the internet without a computer science education.

Henry Sampson: Invented a binder system for propellants and explosives. Co-invented the gamma electric cell, which converts nuclear energy into electricity to power mobile communications, satellites, and space travel. Awarded several patents.

Jesse Eugene Russell: Father of 2G communications, cellular technology, and awarded multiple patents for mobile technology.

Mark Dean: Engineered the ISA systems bus and the GHz RISC processor chip.

Gerald A. Lawson: Video game pioneer.

John Henry Thompson: Built the Lingo programming language and XObjects.

Marc Regis Hannah: Conceptualized 3-D special effects graphics technology used in films.

Valerie Thomas: Innovated 3-D and invented the illusion transmitter used by NASA.

George R. Carruthers: Designed the first observatory on the moon, the far ultraviolet camera/spectrograph, used in the Apollo 16 mission.

Roscoe L. Koontz: Invented the collimator and pinhole gamma-ray camera.

James Edward West: Developed the electroacoustic transducer electret microphone.

David Blackwell: Co-created the Rao–Blackwell theorem, foundational to modern machine learning algorithms.

Clarence "Skip" Ellis: The first Black man to earn a Ph.D in computer science and innovator in operational transformation and collaborative systems, which enabled communication in distributed AI systems.

Roy Clay Sr.: The *"Father of Silicon Valley,"* a lead developer for Hewlett-Packard's first minicomputer and creator of software languages used to evolve modern computing power.

Dr. Nashlie Sephus: Designed visual recognition algorithms for PartPick, which allowed AI to identify mechanical parts from photos.

Dr. Daniel Hale Williams: Heart surgery pioneer.

Patricia Era Bath: Built LASERPHACO for laser eye surgery.

Otis Frank Boykin: Engineered resistors for guided missiles and control units in pacemakers.

Ernest Everett Just: Marine biology and zoology groundbreaker.

Charles Richard Drew: Originated blood transfusions and blood banks.

Garrett Augustus Morgan: Invented the three-position traffic signal and the gas mask.

Benjamin Banneker: Astronomer, mathematician, surveyor, naturalist, and almanac author.

Alexander Miles: Drafted blueprints for modern-day elevators.

Granville Tailer Woods: Created the multiplex telegraph and railway devices.

Benjamin Thornton: Evolved the predecessor to the answering machine device.

Philip B. Downing: Developed an electrical railway switch and the modern-day street postal mailbox.

Henry Brown: Assembled a strongbox for personal valuables.

Charles B. Brooks: Improved the street sweeper.

David N. Crosthwait Jr.: Built systems to heat large structures, the vacuum pump, thermostat control, and the automobile turn signal.

Frederick McKinley Jones: Engineered the portable refrigeration machine, a unit for military air conditioning, a self-start gas engine, and movie projector devices.

Booker T. Whatley: Tuskegee University professor, horticulturist, and author.

Henry Blair: Constructed the corn planter and a mechanical cotton planter.

Thomas Elkins: An abolitionist, dentist, surgeon, and pharmacist who improved the refrigeration apparatus and the modern toilet.

Dr. George Washington Carver: Agricultural scientist, innovated cosmetics, dyes, paints, plastics, cooking oils, peanut products, soil regeneration, crop rotation, composting, and originated the "*Jesup Wagon*," a mobile classroom.

Augustus Jackson: Perfected ice cream preparation and production techniques as a White House chef.

George Speck/Crum: Chef, restaurateur, fried potato experiments led to "*Saratoga chips*."

Nathan "Nearest" Green: Master distiller, mentored Jack Daniel, created Uncle Nearest Premium Whiskey.

Percy Lavon Julian: Researcher, chemist, and originated the chemical synthesis of medicinal drugs from plants.

Shirley Ann Jackson: Telecommunications research led to the touch-tone telephone, portable fax, fiber optic cable, solar cells, call waiting, and caller ID.

Charles Page: Born enslaved in Louisiana, c. 1,864 CE, Page patented an airship, c. 1,906 CE, #US817442A. Page watched bugs fly and said, "*I could do that*."

Thomas Fuller: The Virginia Calculator

NASA Engineers, Mathematicians, Scientists, and Astronauts:

Mae Jemison—the first Black woman in outer space—Katherine Johnson—calculated the moon landing—and engineers: Dorothy Vaughan, Mary Jackson, Jeanette Scissum, Melba Roy, Annie Easley, Ronald McNair, Fred Gregory, Robert Satcher, Leland D. Melvin, Stephanie Wilson, Joan Higginbotham, Michael P. Anderson, Guion S. Bluford Jr., Charles F. Bolden, Yvonne Cagle, Robert L. Curbeam, Benjamin Alvin Drew, Winston E. Scott, Bernard A. Harris Jr..

C. R. Patterson & Sons Cars: Designed automobiles superior to Henry Ford's mass-produced Model T.

Born enslaved, c. 1,833 CE, Charles Richard Patterson escaped a Virginia plantation, trained as a blacksmith, and settled in Greenfield, Ohio. C.R. Patterson & Sons of Greenfield, Ohio, manufactured and repaired carriages, trucks, and buses.

"Our car is made with three distinct purposes in mind. First — It is not intended for a large car. It is designed to take the place originally held by the family surrey. It is a 5-passenger vehicle, ample and luxurious. Second — It is intended to meet the requirements of that class of users, who, though perfectly able to spend twice the amount, yet feel that a machine should not engross a disproportionate share of expenditure, and especially it should not do so to the exclusion of proper provisions for home and home comfort, and the travel of varied other pleasurable and beneficial entertainment. It is a sensibly priced car. Third — It is intended to carry with it (and it does so to perfection) every conceivable convenience and every luxury known to car manufacture. There is absolutely nothing shoddy about it. Nothing skimp and stingy."
-Patterson-Greenfield automobile advertisement listed at \$685.00, c. 1,915 CE.

Handmade vehicles declined as assembly lines revolutionized production. Black American craftsmanship, advocacy through the United Auto Workers, and innovation drove the automotive industry.

History of Low Riders

Low Riders showcase customized paint finishes, murals, windows, hydraulics, body modifications, interiors, engines, drivetrains, suspensions, wheels, rims, and horns.

Black American WWII veterans, skilled in aircraft mechanics, experimented with hydraulics and modified automobiles, c. 1,940-1,980 CE. A notable figure, Sonny Sims, built a powerful multi-pump setup, which enabled cars to lift off the ground or "*hop*," c. 1,967 CE.

Founded by Charles "*Big Red*" Bell, Hot-Rodders established a clubhouse for the *Low Riders*—a name coined by Berta Bell—location 1423 E. 53rd Street, c. 1,952 CE.

The first mention of Low Riders appeared in the article "*Stolen Car Parts Seized, Hot-Rod Gang Arrested.*" -*California Eagle*, Los Angeles, May 30, 1957.

Victor Murphy exhibited adjustable suspension, "*hopping*," craftsmanship installed by Bill Hines of Lynwood, California. Al "*Old Man*" Sullivan produced hydraulic masterpieces from a shop located near Crenshaw Boulevard and 8th Street, Los Angeles, California, c. 1,953 CE.

Athen Nelson engineered "*three-wheel motion*," where one rear wheel lifts while the opposite front and rear wheels remain grounded.

Low Rider Hall of Fame: Ted Wells, Tina L. Blankenship-Early, Gary May...

The band WAR recorded the culture into history with "*Lowrider*," c. 1,975 CE.

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